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Abstract

A steam turbine is a mechanical device that extracts thermal energy from pressurized steam and converts it into rotary motion. A system of angled and shaped blades arranged on a rotor through which steam is passed to generate rotational energy. Moving fluid acts on the blades so that they move and impart rotational energy to the rotor.

The blades are designed in such a way to produce maximum rotational energy by directing the flow of the steam along its surface. The blades are made at specific angles in order to incorporate the net flow of steam over it in its favour. The blades may be of stationary or fixed and rotary or moving types, and shaft is designed to work in extreme conditions, here it has to bear the temperature which is coming from the steam and loads (weight and centrifugal force) of the blades assembly and other assembly parts.

The aim of the project is to design a steam turbine blade and shaft assembly using 3D modelling software Pro/Engineer using the CMM point's data collected from HPCL Vishakhapatnam. And simulating structural, vibrational and thermal analysis on assembly of blade and shaft by applying different materials. By conducting above analysis stresses developing on blade, mode shape of the blade and thermal behaviour are found. Using analysis results the best material for both shaft and blade is suggested.

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DESIGN AND ANALYSIS OF STEAM TURBINE BLADE AND
SHAFT ASSEMBLY



Steam turbine showing blade and shaft assembly

INTRODUCTION

STEAM TURBINE

A turbine is a rotary mechanical device that extracts energy from a fluid flow and converts it into useful work. A turbine is a turbo machine with at least one moving part called a rotor assembly, with blades attached. Moving fluid acts on the blades so that they move and impart rotational energy to the rotor. Early turbine examples are windmills and waterwheels.

The word "turbine" was coined in 1822 by the French mining engineer Claude Burdin from the Latin turbo. Gas, steam and water turbines usually have a casing around the blades that contains and controls the working fluid. Credit for invention of the steam turbine is given both to the British engineer Sir Charles Parsons (1854–1931) for invention of the reaction turbine and to Swedish engineer Gustaf de Laval (1845–1913) for invention of the impulse turbine. Modern steam turbines frequently employ both reaction and impulse in the same unit, typically varying the degree of reaction and impulse from the blade root to its periphery.

STEAM TURBINE

A steam turbine is a device that extracts thermal energy from pressurized steam and uses it to do mechanical work on a rotating output shaft.

Because the turbine generates rotary motion, it is particularly suited to be used to drive an electrical generator and pumps about 90% of all electricity generation in the United States (1996) is by use of steam turbines. The steam turbine is a form of heat engine that derives much of its improvement in thermodynamic efficiency from the use of multiple stages in the expansion of the steam, which results in a closer approach to the ideal reversible expansion process.

The modern steam turbine was invented in 1884 by Sir Charles Parsons, whose first model was connected to a dynamo that generated 7.5 kW (10 Hp) of electricity. The invention of Parson's steam turbine made cheap and plentiful electricity possible and revolutionized marine transport and naval warfare. Parsons' design was a reaction type.

CLASSIFICATION OF STEAM TURBINES BY THE ACTION OF STEAM

1. Impulse
2. Reaction
3. Impulse and reaction combined

THE NUMBER OF STEP REDUCTIONS INVOLVED

1. Single stage
2. Multi-stage

THE DIRECTION OF STEAM FLOW

1. Axial
2. Radial
3. Mixed
4. Tangential

THE INLET STEAM PRESSURE

1. High pressure
2. Medium pressure
3. Low pressure

THE FINAL PRESSURE

1. Condensing
2. Non-condensing

THE SOURCE OF STEAM

3. Extraction
4. reheat

INTRODUCTION TO HPCL

Hindustan Petroleum Corporation Ltd is a mega public sector undertaking (PSU) and is the second largest integrated oil company with Navarathna status. HPCL represents 25% of the country's oil capacity. Visakha Refinery was established in 1957 as CALTEX OIL REFINING INDIA LIMITED (CORIL). This was the first oil refinery on the east coast and the first major industry in the city of Visakhapatnam. Hindustan Petroleum Corporation came into being in mid-1974 after take over and merging of Erstwhile Esso and Lube India in 1976 and was subsequently merged with HPCL Kosan Gas Company in 1978. HPCL thus came into being after merging four different organizations at different parts of time.

Initial installed capacity of 0.675 MMTPA in 1957. The crude processing capacity is raised to 8.3 MMTPA throughout level over a period of years by adding various Units and modifications. But, various modifications and efficient productivity helped refinery to achieve 9.0 MMTPA for consecutive 3 years. Refinery is capable of processing both imported & indigenous Crude's. Refinery has processed various types of Bituminous and Non-Bituminous Crude's since its inception. DHDS (Diesel Hydro-Desulphurization) and related utilities/offsite facilities are added for enhancing the quality of diesel product to meet Environment norms. Similarly, MS (motor spirit) block and related utilities/offsite facilities are added for enhancing the quality of Petrol product.

VR has its additional storage facilities at the North of the refinery called as additional Tankage Project (ATP). The ATP storage tanks are spread over area of 215 acres.



Image of Steam Turbine in HPCL VISHAKAPATANAM.

MODELING AND ANALYSIS

GENERATION OF PRESSURE DISTRIBUTION DATA ON THE BLADE



SURFACE

Last stage blade of steam turbine, which is being analysed for stress and vibration is a highly twisted blade due to the variation if the blade speeds across the height of the blade. The deflection in the blade passage also reduces from hub to tip to vary the loading on each section. Thus the pressure distribution on the suction and pressure surface of the blade changes considerably from hub to tip to match the loading at that suction. It is known fact that the area of pressure distribution curve representing the blade loading. Hence it has been decided to generate the pressure distribution at all the '11' blade sections.

The following procedure is allows to get the blade surface pressure distribution with the help of BladeGen and BladeGen plus package.

1.From the blade coordinate input data file for suction/pressure surface x, y, z coordinate of surface was generated as a loop with the following notations.

X-along the height of the blade.

Y- Meridional direction.

Z-along blade to blade

2.Profile curve is generated with above coordinates of all sections placed one below the other is sequence from section (1) to section (5) along the height of the blade. The coordinates between two sections are separated.

3.Hub & Shroud boundary is generated at the appropriate heights with -Y negative meridional axis corresponded from LE (Leading edge). And positive distance from meridional distance from TE (Tailing Edge).

4.Hub Curve file is generated as follows:

X, Y, Z

```
283.450000  0.000000000  -100.000000
283.450000  0.000000000   0.000000000
283.450000  0.000000000  100.000000
```

In between the values Comma is compulsory. (X, Y, Z)

A profile contains total 60 points for all sections.

5.Profile Curve file is generated as follows:

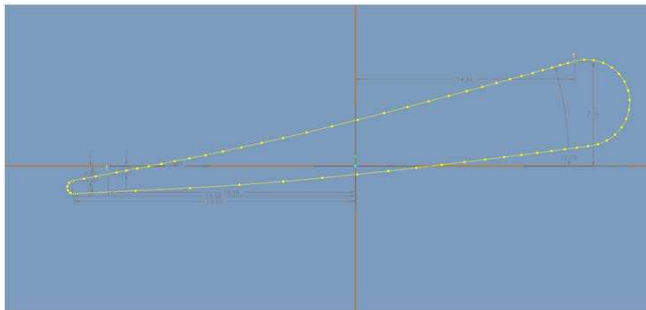
X, Y, Z

#

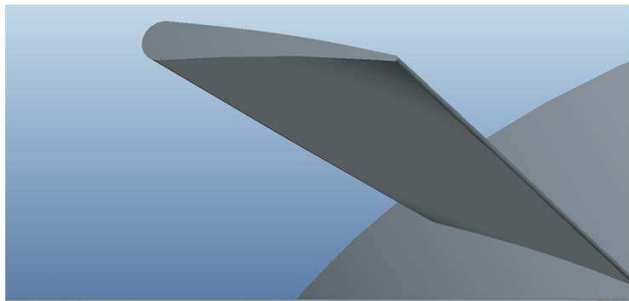
```
283.45,-5.74,-22.92
283.45,-5.23,-23.25
283.45,-4.46,-23.36
```

283.45,-3.43,-23.22
 283.45,-2.15,-22.82
 283.45,-0.66,-22.12
 283.45, 1.03,-21.11
 283.45, 2.85,-19.72
 283.45, 4.74,-17.91
 283.45, 6.61,-15.62
 283.45, 8.32,-12.78
 283.45, 9.66,-9.39
 283.45, 10.4,-5.53
 283.45, 10.35,-1.43
 #442.65, 15.21,-15.51
 442.65, 15.64,-15.21
 442.65, 15.81,-14.69
 442.65, 15.74,-13.95
 442.65, 15.44,-12.99
 442.65, 14.91,-11.83
 442.65, 14.19,-10.49
 442.65, 13.26,-8.99
 442.65, 12.14,-7.36
 442.65, 10.79,-5.66
 442.65, 9.23,-3.95

BLADE PROFILE



Turbine blade profile in Pro/E Sketcher.



Generated part of Blade

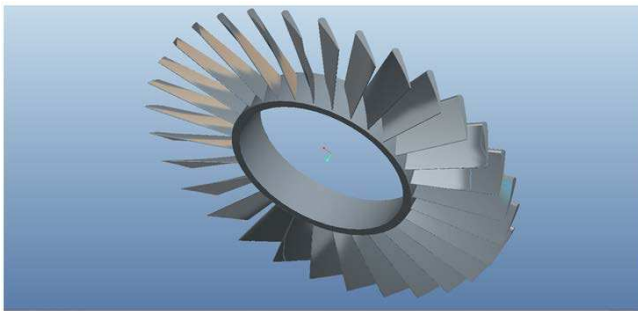
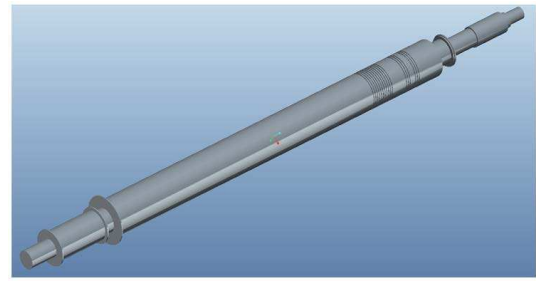


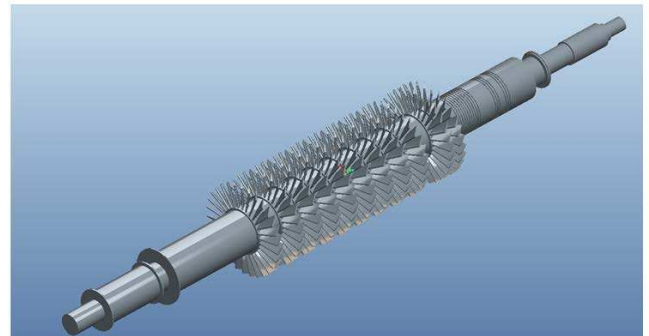
Image showing blade assembly.



Generated part of shaft.

Step by steps procedure for Assembly:

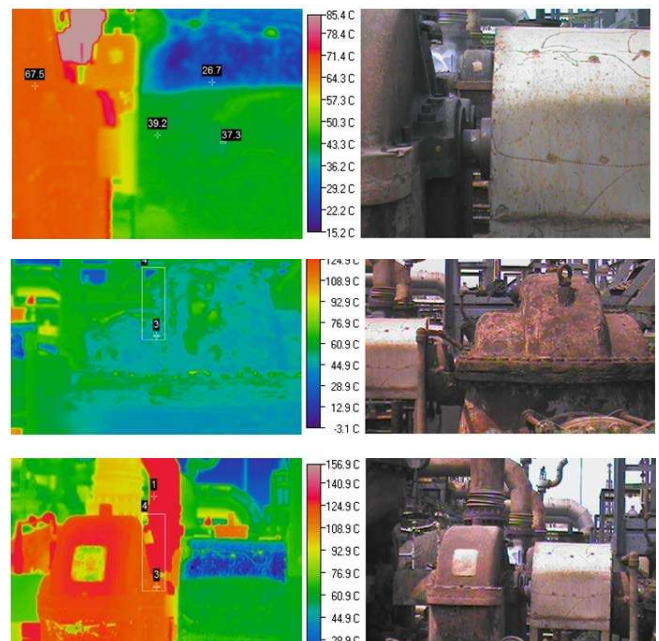
- After opening new assembly file shaft is assembled with default constraints.
- Patterned blade part is assembled using insert, mate and align constraints at required positions
- File is exported to IGES(Initial Graphical Exchanging Specification) for analysis purpose



Turbine blade and shaft assembly

Now using these design conditions and working conditions of the turbine. The structural, vibrational and thermal analysis is carried out with three different materials as mentioned.

The temperature changes in the turbine are monitored periodically using the thermal images taken by the thermal camera. When a image taken by the thermal camera the image will show the temperature variations directly



INTRODUCTION TO FEA

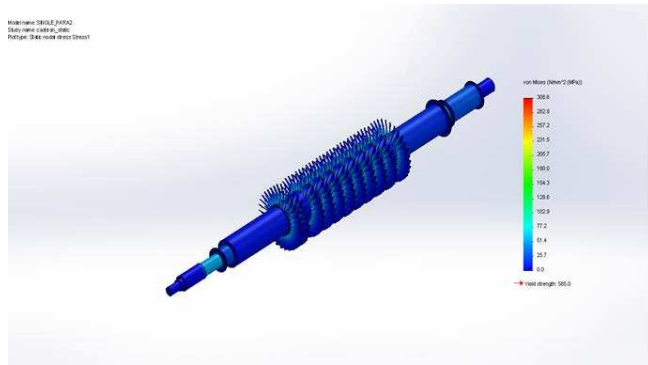
Finite Element Analysis (FEA) was first developed in 1943 by R. Courant, who utilized the Ritz method of numerical analysis and minimization of variational calculus to obtain approximate solutions to vibration systems. Shortly thereafter, a paper published in 1956 by M. J. Turner, R. W. Clough, H. C. Martin, and L. J. Topp established a broader definition of numerical analysis. The paper centered on the "stiffness and

RESULTS AND DISCUSSION

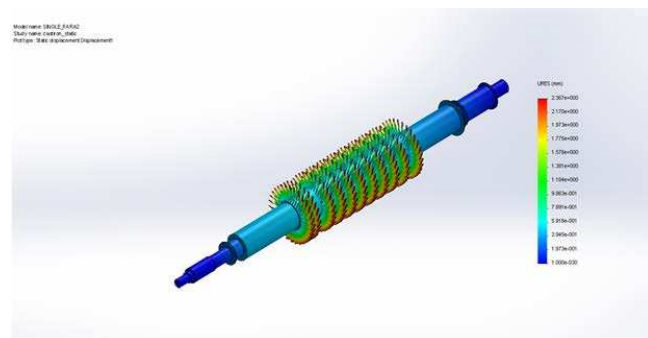
When the simulation is completed a report is generated in order to explore the results. At first the static analysis is done for three different types of materials. It gives stresses developed in the turbine blade and shaft assembly, strain values and displacement.

Now all the stress, strain and displacement values are shown in figures below.

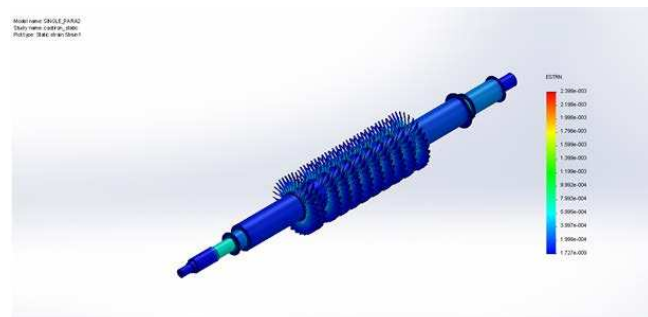
RESULTS OF STRUCTURAL ANALYSIS



Stress developed in EN24 Stainless Steel

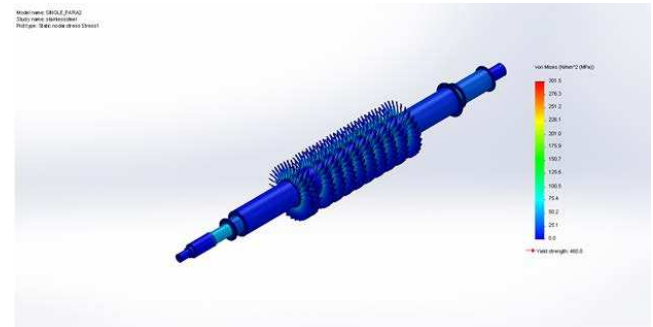


Displacement of EN24 Stainless Steel.

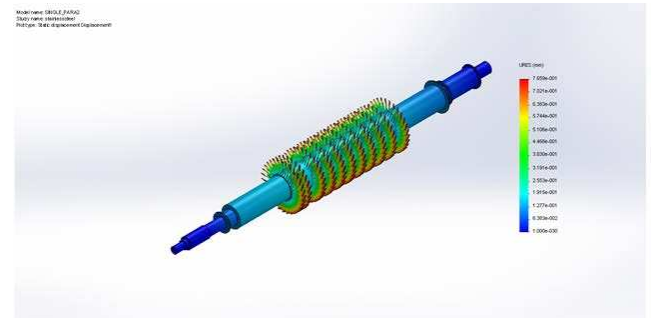


Strain for EN24 Stainless Steel.

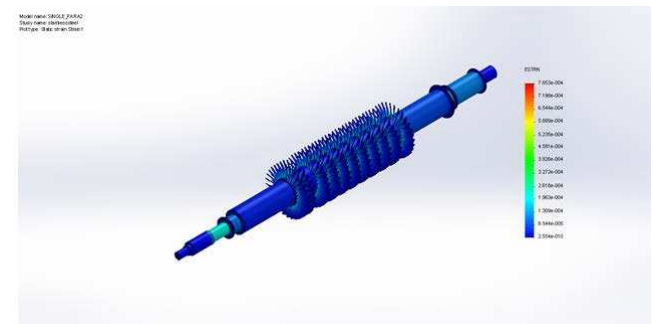
AISI 4130 Steel



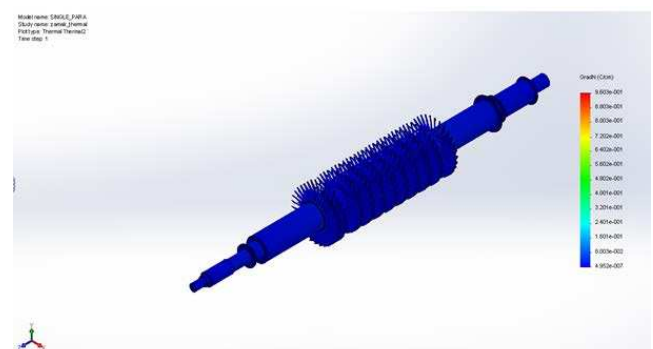
Stress developed in AISI 4130 Steel



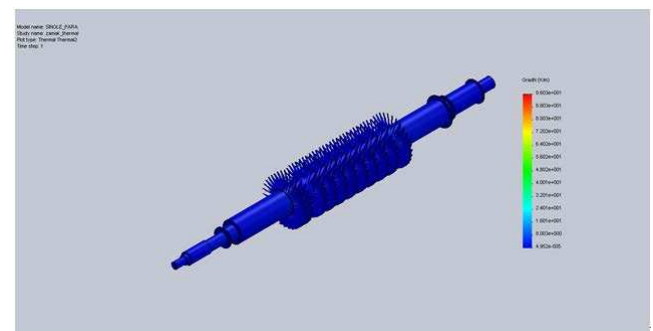
Displacement of AISI 4130 Steel



Strain for AISI 4130 Steel.



Thermal gradient for ZAMAK

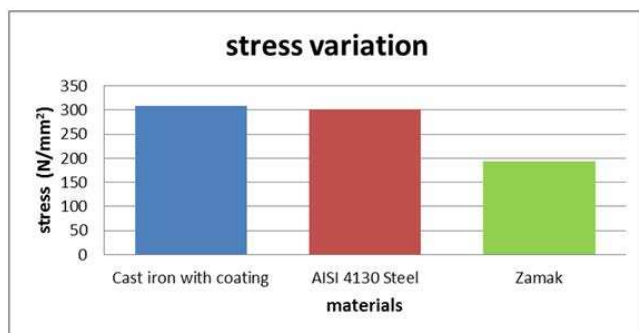


Thermal flux for ZAMAK.

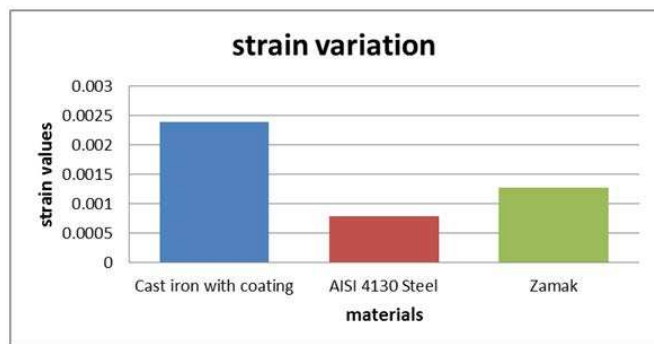
RESULT TABLES AND GRAPHS

After the loads are applied to the imported model, the meshing is done successfully and the results are compiled. At the static analysis is done for three materials EN24 Stainless Steel, AISI 4130 Steel, ZAMAK. The results are tabulated below:

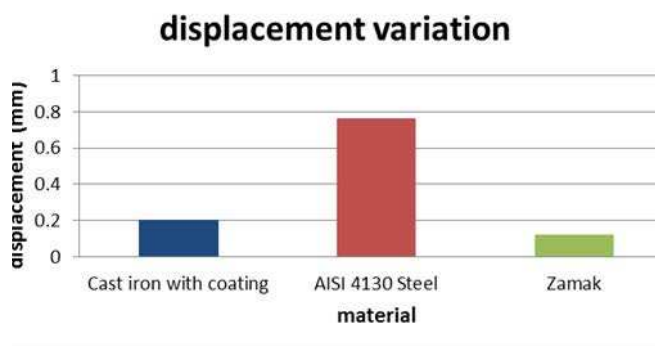
S.No	Material	Stress N/mm ²	Strain	Displacement mm
1	EN24 Stainless Steel	308.601	0.00239799	0.2036723
2	AISI 4130 Steel	301.458	0.000785258	0.765902
3	ZAMAK	192.972	0.00127763	0.123308



Graph showing stress variation for all the three materials



Graph showing strain variation for all the three materials.



Graph showing displacement variation for all the three materials

After successful completion of static analysis again the model is imported in to the Cosmos taking a new study. The model is solid meshed and appropriate constraints are specified to run simulation. In vibrational analysis the results are plotted for five mode shapes which gives different displacement values at different frequencies. The

results are tabulated as shown below.

Result summary for vibrational analysis for EN24 Stainless Steel

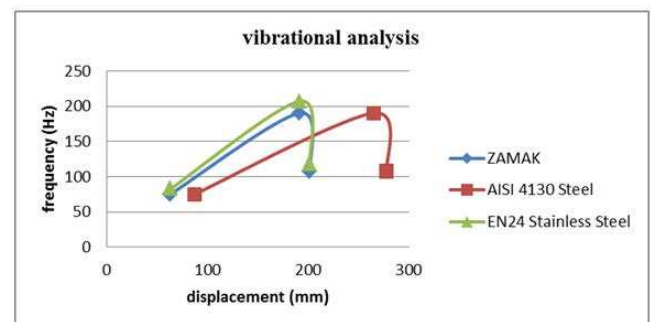
S No	Mode shape1	Mode shape2	Mode shape3	Mode shape4	Mode shape5
Fre-quency	62.962	62.977	191.11	201.52	201.57
dis- placement	82.4979	82.4925	207.316	118.441	118.893

Result summary for vibrational analysis for AISI 4130 Steel

S No	Mode shape1	Mode shape2	Mode shape3	Mode shape4	Mode shape5
Fre-quency	86.767	86.787	264.76	277.72	277.78
dis- placement	72.3369	72.3447	182.487	103.784	104.379

Result summary for vibrational analysis for ZAMAK

S No	Mode shape1	Mode shape2	Mode shape3	Mode shape4	Mode shape5
Fre-quency	62.962	62.977	191.11	201.52	201.57
dis- placement	74.9789	74.9846	190.379	107.706	108.248

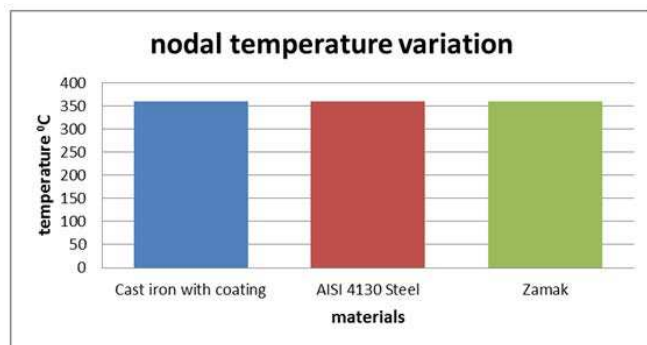


Graph showing different mode shape curves for all the three materials.

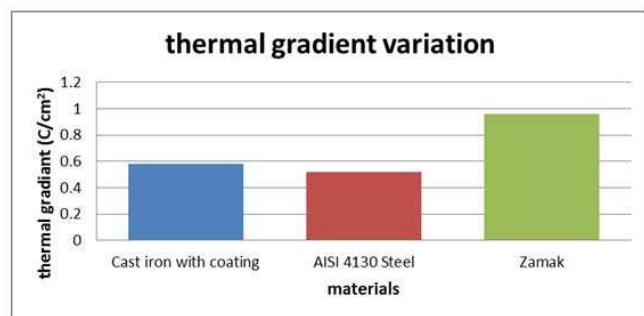
Now the thermal analysis is carried out by importing the material and meshing is done with solid mesh. The constraints are applied to blades with an inlet temperature of 3600 C. the results are tabulated below.

Result summary table for thermal analysis

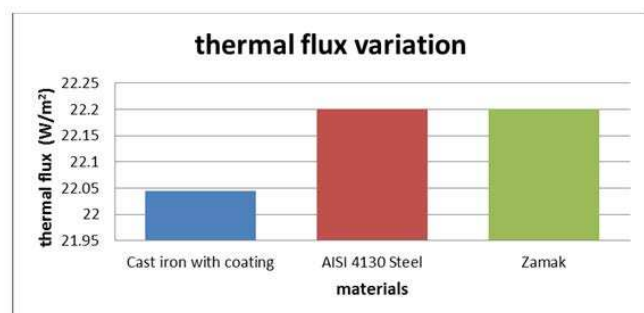
S.No	Material	Nodal temperature Celsius	Thermal gradient	Thermal flux W/m ²
1	EN24 Stainless Steel	360	0.580106	22.044
2	AISI 4130 Steel	360	0.519904	22.1999
3	ZAMAK	360	0.960332	22.1999



Graph showing nodal temperature variation for all the three materials



Graph showing thermal gradient variation for all the three materials



Graph showing thermal flux variation for all the three materials

CONCLUSIONS

The entire project work is done in R&D department of HPCL Visakhapatnam for optimizing the material of steam turbine assembly.

A PT2001 turbodine steam turbine is optimized for to reduce maintenance. Initially static and thermal conditions are evaluated using Infra-red thermometer and digital vibrometer. Those readings are taken for simulation inputs A FEA model is developed according to given drawing. Static analysis is carried out on FE model using EN24 Stainless steel (present material), AISI 4130 Stainless Steel and zinc aluminum alloy (zamak) with zirconia coating.

In static analysis, the stress value of ZAMAK is best when compared with other materials and value is 192.972 N/mm². The strain value of AISI 4130 Steel is best with a value of 0.000785258. the displacement for ZAMAK is 0.123308.

Vibrational analysis is carried out to determine the vibrations due to geometry and property of material.

In vibrational analysis, ZAMAK is having less displacement at a particular frequencies among all the three materials as shown in the table 5.5

Thermal analysis is carried out to determine the thermal

behavior like thermal gradient and heat flux.

In thermal analysis, ZAMAK is having high thermal gradient and thermal flux and the thermal gradient is 0.960332 °C/cm and thermal flux is 22.1999 W/m².

Partially stabilized zirconia is mainly used as a surface coating to prevent the thermal effect on surface and also it reduces the corrosive effect.

As per the analytical results ZAMAK material along with partially stabilized zirconia coating will improve reliability of turbine shaft and blades due to less stress, negligible displacement and strain values, also ZAMAK is having good level of thermal gradient(heat transfer rate) and sufficient heat flux rate which in turn improves the power generation rate by reducing the maintenance.

SCOPE OF FUTURE WORK

In this project the total work is carried out to design and analysis of steam turbine blade and shaft assembly by applying different materials and finally suggesting best material for blade and shaft. The future scope of this project would be designing of turbine blade with different angles and shapes to get more outputs. The new materials can be applied to blade and shaft assembly which can reduce maintenance cost and greater power generation without loss.

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